

Sankalp Rajeev

Dearborn, MI | (480) 208-2139 | srajeev@umich.edu | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

University of Michigan – Dearborn

M.Sc., Artificial Intelligence & M.S.E., Robotics Engineering; GPA: 3.90/4.00

Sep 2024 – Dec 2026 (Expected)

Dearborn, MI

Arizona State University

B.S., Computer Science; Minor: Mathematical Concepts of Engineering; GPA: 3.71/4.00

Aug 2020 – May 2024

Tempe, AZ

WORK EXPERIENCE

Zoetis

AI Engineering Intern

May 2025 – Nov 2025

Kalamazoo, MI

- Deployed 4 production LLM tools (GPT-4, Claude) on Azure Kubernetes, cutting regulated documentation workload by 1,000+ engineer-hours annually via async ETL and semantic similarity pipelines for ServiceNow workflows.
- Delivered a real-time SQL-backed analytics dashboard to track LLM tool adoption, reducing incident misclassification by ~40% across cross-functional teams.

Mistral Solutions Pvt Ltd

AI/ML Engineer Intern

Jun 2024 – Aug 2024

Bangalore, India

- Deployed YOLOv8 on Qualcomm QRB5165 for real-time UAV object detection; applied SNPE SDK mixed-precision quantization, reducing inference latency by 25%.

Astroseed – Terra Pod Capstone

Software Engineer (Scrum Master)

Aug 2023 – May 2024

Tempe, AZ

- Led autonomous robotic arm development integrating YOLOv5 (94% accuracy) with depth sensing and ROS-based SLAM in Gazebo; coordinated Agile sprints as Scrum Master.

Machani Robotics

Software Engineering Intern

Jun 2022 – Aug 2022

Bangalore, India

- Developed control software for a 6-DOF robotic arm via CAN protocol (C++/Python) with an integrated ROS vision feedback pipeline.

TECHNICAL PROJECTS

Kairos – Demand Intelligence Platform | PySpark, Kafka, Airflow, PyTorch, XGBoost, FastAPI, Docker

- Designed a production data lake ingesting **411M+ e-commerce events** (14GB, REES46) using PySpark batch ETL orchestrated by **Apache Airflow DAGs** and Kafka Structured Streaming (5-min tumbling windows) into **TimescaleDB**, producing 3 ML-ready feature tables with AQE and skew-join optimization.
- Built a three-stage inference funnel – **SASRec Transformer** (NDCG@10 $\approx$ 0.90) retrieves 200 candidates in 1.5ms, Two-Tower FP16 re-scores, XGBoost ranker (**AUC 0.9883**) ranks by purchase probability – with **PgBouncer** multiplexing 120 connections to 25 for sub-10ms latency.
- Served via FastAPI + **Redis** cache; instrumented with **Prometheus/Grafana** (p50/p95/p99); 17-service Docker Compose with GitHub Actions CI/CD (83 Pytest tests, Ruff zero-warning); XGBoost forecaster achieved **WMAPE 9.44%**.

CodeMind – AI Code Intelligence Platform | Python, CrewAI, RAG, MLflow, Ollama

- Architected a local-first **multi-agent system** (CrewAI, Ollama) with sequential, parallel, and iterative crew patterns; built **hybrid RAG** (CodeBERT + BM25 + RRF) with AST-based semantic chunking achieving **15.4ms retrieval latency** on 42K+ indexed chunks; instrumented with **MLflow** and anti-hallucination prompting via chain-of-verification.
- Fine-tuned Gemma 3 (12B) on the **KodCode dataset** using **Unsloth QLoRA** (4-bit Quantization, A100); evaluated with **DeepEval** using **Gemini-2.0-Flash** as **LLM-as-a-Judge**, achieving **73.5% CodeBLEU improvement**, 96% assertion density, and 3.3 $\times$ inference speedup.

MeetingMind AI – Multimodal Meeting Intelligence | Python, FastAPI, React, Docker, GCP

- Architected a 5-phase **multimodal ML pipeline** using FFmpeg, **Whisper large-v3** (<10% WER), **Pyannote** diarization, YOLOv11, and **Vertex AI Gemini** to produce speaker-attributed transcripts with action items; parallelized phases via **ThreadPoolExecutor** for end-to-end throughput.
- Reduced **VLM API calls by 85%** using a multi-signal content pre-filter (SSIM scene change detection, EasyOCR text density gate, skin-tone camera filter) to restrict analysis to high-value semantic frames; matched speakers to faces via **IoU tracking** and 512-dim **ArcFace** cosine similarity.
- Deployed **FastAPI** backend on **Google Cloud Run** with **Redis Queue** for async job processing and **ChromaDB** + **LangChain** RAG for meeting Q&A; built React 18 frontend with real-time status polling and video-transcript sync.

SKILLS

Languages

Python, C/C++, SQL, TypeScript, Java

ML & AI

PyTorch, Transformers, XGBoost, scikit-learn, HuggingFace, Whisper, Pyannote, QLoRA

Data Engineering

PySpark, Kafka, Airflow (DAGs), TimescaleDB, Redis, PgBouncer, Polars, Pandas, MinIO

LLMs & Agents

RAG (Hybrid/RRF), CrewAI, LangChain/LangGraph, LLM APIs (GPT-4, Claude, Gemini)

Cloud & DevOps

Docker, GitHub Actions, GCP (Cloud Run, Vertex AI), Azure (AKS), FastAPI, Prometheus, Grafana

Computer Vision

YOLO (v5/v8/v11), OpenCV, InsightFace, EasyOCR, SLAM, ROS